J Head Trauma Rehabil
Vol. 00, No. 0, pp. 1–10
Copyright © 2026 The Authors. Published by Wolters Kluwer Health, Inc.

OPEN

Does Timing of Concussion in Season Affect Incidence and Recovery in Division I Collegiate Football Players? A CARE Consortium Study

Anthony E. Bishay, BS; Samuel W. Fitch; Chibuikem S. Iheagwaram, BS; Kristen L. Williams, MS, LAT, ATC; Christopher Naranjo, BS; Tyler K. Sills, MOL; Robert W. Fitch, MD; Paul F. Pasquina, MD; Thomas W. McAllister, MD; Michael A. McCrea, PhD, ABPP; Steven P. Broglio, PhD, ATC; Douglas P. Terry, PhD; Scott L. Zuckerman, MD, MPH, CARE Consortium Investigators

Objective: To investigate whether seasonality is associated with (1) concussion incidence and (2) recovery outcomes, including the time to symptom resolution (SR) and return-to-play (RTP) in Division I college football players. **Setting:** Multi-institution study using the CARE Consortium database composed of 30 universities investigating the natural history of concussions between 2014 and 2020. **Participants:** 682 Division I football athletes across 21 institutions who sustained a concussion during practice or competition. **Design:** Prospective cohort study of NCAA Division I football players with concussion. **Main Measures:** Concussion incidence rates across 4 season periods (preseason, early, middle, and late-season) were analyzed using chi-square tests. Time to SR and RTP were analyzed using Kaplan-Meier curves, and Cox proportional hazards regression. **Results:** Concussion incidence rates were highest during preseason (4.73 per 100 players) compared to early (2.19), middle (2.31), and late-season (2.17). The temporal distribution showed 41.9% of concussions occurred during preseason, with 19.2%, 20.1%, and 18.8% occurring during early, middle, and late-season, respectively ($\chi^2 = 104.6$, $P < .001$). Kaplan-Meier analysis revealed significant differences in time to SR (log-rank $P = .012$). Early (HR = 0.582, $P < .001$) and late-season (HR = 0.634, $P = .002$) concussions had delayed SR compared to preseason. No differences in time to RTP (log-rank $P = .300$) were found, although early-season concussions had a lower RTP hazard (HR = 0.703, $P = .040$). **Conclusions:** In a multi-institution cohort of collegiate football players, preseason resulted in the highest incidence of concussions, while early and late-season concussions had significantly longer time to SR compared to preseason. These temporal patterns suggest that contextual factors may influence concussion outcomes and warrant mechanistic investigation to inform future management strategies. **Key words:** *American football, return-to-play, seasonality, symptom resolution*

Author Affiliations: *Vanderbilt University School of Medicine, Nashville, Tennessee (Mr Bishay and Mr Naranjo); Vanderbilt Sports Concussion Center, Vanderbilt University Medical Center, Nashville, Tennessee (Mr Bishay, Mr Fitch, Mr Naranjo, Mr Iheagwaram, Dr Terry, and Dr Zuckerman); Vanderbilt University, Nashville, Tennessee (Mr Fitch); Department of Neurological Surgery, Vanderbilt University Medical Center, Nashville, Tennessee (Mrs Williams, Drs Terry and Zuckerman); Department of Orthopedic Surgery, Vanderbilt University Medical Center, Nashville, Tennessee (Dr Fitch); Department of Physical Medicine and Rehabilitation, The Uniformed Services University of the Health Sciences in Bethesda, Bethesda, MD (Dr Pasquina); Department of Psychiatry, Indiana University School of Medicine, Indianapolis, IN (Dr McAllister); Department of Neurosurgery, Medical College of Wisconsin, Milwaukee, WI (Dr McCrea); Department of Neurology, Medical College of Wisconsin, Milwaukee, WI (Dr McCrea); and Michigan Concussion Center, University of Michigan, Ann Arbor, MI (Dr Broglio).*

CARE Consortium Investigators are listed alphabetically by last name: Holly J. Benjamin, MD (University of Chicago), Kenneth L. Cameron, PhD (United States Military Academy), James R Clugston, MD (University of Florida), Stefan M. Duma, PhD (Virginia Tech), James T. Eckner, MD, MS (University of Michigan), Luis A. Feigenbaum, PT, DPT, ATC, LAT (University of Miami), Joshua T. Goldman, MD (University of California, Los Angeles), Joseph B. Hazzard, Jr, ATC (University of Houston-Clear Lake), Thomas W. Kaminski, PhD (University of Delaware), Louise A. Kelly, PhD (California Lutheran University), Anthony P. Kontos, PhD (University of Pittsburgh Medical Center), Christina L. Master, MD (University of Pennsylvania Perelman School of Medicine), Christopher M. Miles, MD (Wake Forest University), Justus Ortega, PhD (California State Polytechnic University, Humboldt), Nicholas Port, PhD (Indiana University), Margot Putukian, MD (Major League Soccer), and Steve Rowson, PhD (Virginia Tech).

This publication was made possible in part by the National Collegiate Athletic Association (NCAA) and the Department of Defense (DOD). The U.S. Army Medical Research Acquisition Activity is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Combat Casualty Care Research Program, endorsed by the Department of Defense through the Joint Program Committee 6/Combat Casualty Care Research Program–Psychological Health and Traumatic Brain Injury Program under Award Nos. W81XWH1420151 and W81XWH1820047. Opinions, interpretations, conclusions, and recommendations are those of the author and are not necessarily endorsed by the Department of Defense.

P.F.P. is an employee of the DoD and has previously received other DoD-related grants; T.W.M. has received research support from the NCAA, the

A CONCUSSION IS the mildest form of traumatic brain injury and occurs commonly in collegiate athletics.[1,2] An estimated 300,000 sport-related concussions occur in the US annually,[1] and previous studies have suggested that collegiate athletes who engage in high-contact sports, such as football, wrestling, and rugby, are more prone to sustaining a concussion. Men's football appears to have the highest amount of sport-related concussion occurrences per year (n = 3,417) at a rate of 6.71 per 10,000 athletic exposures.[3] Investigating the factors leading to differing concussion risks and outcomes is paramount to creating safer gameplay and better management. Multiple studies have assessed risk factors for a prolonged recovery after concussion such as demographics, medical history, acute injury signs/characteristics, symptom severity, the type of competition, and mechanism of the injury.[4-12]

More recently, researchers have examined whether seasonality (ie, the time point in the season when a concussion occurs) is associated with outcomes of concussion.[13-15] For instance, in a retrospective study of athletes across multiple concussion centers, Brett et al[13] found higher initial total symptom scores for high-school football athletes who sustained a concussion later in the season compared to athletes who sustained a concussion early in the season. Additionally, a recent study of high school football athletes who sought care at a specialty concussion clinic found that middle-season concussions occurred most often, and those who sustained a late-season concussion took longer to recover compared to those who sustained a concussion earlier in the season.[14] At the collegiate level, a study of 1120 Division 1 football athletes demonstrated that elevated preseason head impact exposure was associated with increased concussion vulnerability persisting throughout the entire fall season.[16] Similarly, an investigation of 19 NCAA Division I football programs revealed significantly higher concussion rates and head impact exposures during both spring season and preseason practices compared to regular-season.[17] However, prior studies have not examined whether concussion incidence and recovery vary across specific phases within the regular-season itself. Previous work either compared only preseason vs. regular-season as broad categories[16,17] or examined high school populations using retrospective designs.[13,15] Whether within-season patterns exist in a large, prospectively collected collegiate cohort remains unknown. Additionally, it remains unclear whether temporal patterns in recovery persist after accounting for established predictors such as symptom severity and injury characteristics. The regular-season spans months during which contextual factors vary considerably. For instance, early-season games occur as athletes establish starting positions, mid-season represents peak competitive pressures, and late-season carries playoff implications. These shifting contexts may influence both incidence (through changes in play intensity or cumulative exposure) and recovery (through varying pressures to return or academic demands). Identifying such patterns is a necessary first step before designing mechanistic studies to test specific hypotheses about the factors driving temporal variation.

Given the prior associations of seasonality with both the incidence and acute symptomatology after sport-related concussion in high school athletes, we chose to further evaluate seasonality and concussion at the Division I collegiate level. In a multi-institution sample of collegiate football athletes,[18] we sought to examine whether concussion incidence and recovery outcomes, including time to symptom resolution (SR) and return-to-play (RTP), vary across season phases.

METHODS

Participants

As a part of the Concussion Assessment, Research, and Education (CARE) Consortium, 30 universities engaged in a multisite prospective cohort study investigating the natural history of concussion.[18] CARE baseline and post-concussion data between 2014 and 2020 (6 seasons) were used in the current study. Prior to data

US DoD, and the National Institutes of Health (NIH). He also reports textbook royalties from American Psychiatric Press Publishing Inc and serves on the Concussion Committee Scientific Advisory Board for the Australian Football League; M.A.M. has received research funding from the NIH, the Centers for Disease Control and Prevention (CDC), the DoD, the NCAA, the National Football League (NFL), and Abbott Laboratories; and Steven P. Broglio has received research funding from the NIH, the CDC, the DoD-USA Medical Research Acquisition Activity, the NCAA, the NATA Research and Education Foundation, the NFL/Under Armour/GE, Simbex, and ElmindA. He has consulted for US Soccer, US Cycling, the University of Calgary SHRed Concussions external advisory board, and medicolegal litigation; S.L.Z. reports being an unaffiliated neurotrauma consultant for the NFL and a consultant for Medtronic; D.T. receives funding from Amgen, Inc., and Football Research, Inc. He is a consultant for the NFL as their Senior Director of Research. He also serves as a scientific advisor for HitIQ. He previously consulted for REACT Neuro, Inc. He has a consulting practice in forensic neuropsychology, including expert testimony, involving individuals who have sustained mild TBIs (including former athletes). The views expressed in this article are those of the authors and do not reflect the official policy or position of the US government, DoD (Defense Health Program funds), or NFL.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's website (www.headtraumarehab.com).

This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

Corresponding Author: *Scott L. Zuckerman, MD, MPH, Vanderbilt Sports Concussion Center, Department of Neurological Surgery, Vanderbilt University Medical Center, 1211 Medical Center Drive, Nashville, TN 37212 (scott.zuckerman@vumc.org).*

DOI: 10.1097/HTR.0000000000001146

collection, each institution's local institutional review board and the Human Research Protection Office approved the study protocol. All participants provided written informed consent. For this study, only football players across 21 institutions in the major NCAA Division I football conferences (Supplemental Digital Content Table 1, available at: http://links.lww.com/JHTR/B25) who had baseline data and who sustained a concussion during the preseason and regular-season were included. Division I conferences were selected due to their comparable level of support, which differs from Division II and III programs. Concussions that occurred after the regular-season ended (ie, during the playoffs) or at other points of the year were not analyzed since only 4 teams competed during playoffs during the study period.[19] No further inclusion or exclusion criteria were applied.

Data collection

As previously reported,[20] participants at each institution completed a pre-season baseline evaluation. Self-reported information collected at baseline included sex, race, ethnicity, history of pre-existing conditions (eg, learning disability [LD], Attention-Deficit/Hyperactivity Disorder (ADHD), migraine headaches, and psychological disorders), and baseline Sport Concussion Assessment Tool-3rd Edition Total Symptom Severity (SCAT3 TSS) scores. Data were collected between 2014 and 2020, during which the SCAT3 TSS was used to ensure consistency in data collection. Following concussion, participants completed post-injury SCAT3 assessments within 24 hours of injury. Psychological disorders included depression, other moods disorders, anxiety disorders, substance use disorders, personality disorders, psychotic-spectrum disorders, and Post-Traumatic Stress Disorder. All sites used a common definition of concussion,[21] and concussions were diagnosed clinically by team medical staff.

Independent variable

The primary independent variable examined was *seasonality*, which was operationally defined as the timing of concussions within a sport season. We use the term "seasonality" consistent with prior concussion literature examining temporal patterns across different phases of the competitive season.[13-15] Specifically, each concussion was categorized into 1 of 4 groups: *preseason; early-season; middle-season; late-season*. To determine the boundaries of these 4 groups, the dates of the preseason and regular-season were determined for each individual college in the study for each of the 6 individual seasons, since each year included different season start/end dates (ie, 6 seasons, 21 schools across 10 leagues, yielding a total of 126 individual seasons). Time was categorized into discrete phases rather than analyzed as a continuous variable for several reasons: (1) to facilitate comparison with prior literature that used similar categorical approaches, (2) continuous analysis assumes a linear relationship between timing and recovery, which may obscure complex patterns where recovery times peak or dip at specific points during the season, and (3) discrete season phases are more clinically meaningful and interpretable to practitioners than continuous day-of-season coefficients. For instance, understanding recovery time differences across established phases (eg, early vs. late-season) is more useful to practitioners than interpreting marginal changes associated with each successive day of the season. Preseason was defined as the period from the start of official team practices (typically early August) until the first regular-season game. The preseason date boundaries were based on extracted team data for each institution and season. For early/middle/late-season date boundaries, the total number of days from the start of regular-season to the last game of the regular-season was divided into 3 equal time epochs. Each concussion was then grouped into 1 of the 4 aforementioned categories. Concussions that occurred outside the defined study periods for each team were excluded from analysis. Date breakdowns for each Division I conference are provided in Supplemental Digital Content Table 1, available at: http://links.lww.com/JHTR/B25.

Outcomes

The primary outcome was concussion incidence, defined as the number of new concussions during each season period divided by the total number of athletes at risk during that same period, expressed per 100 players.[22] To more accurately reflect the at-risk population, players were considered at risk if they were on the active roster and not sidelined due to injury. Athletes were removed from the denominator upon injury and re-added following their RTP date. Secondary outcomes included two recovery outcomes of days to SR and RTP. Days to SR was defined as the time from injury to complete SR or return to baseline symptom burden across the CARE sites based on clinical judgment. Days to RTP were defined as the time from injury to unrestricted RTP.

Statistical analysis

Descriptive statistics are presented for demographics, personal/family medical history, and injury characteristics stratified by seasonality groups. Categorical variables were presented as frequencies, while continuous variables were expressed as M ± SD. Pearson's chi-square tests of association were conducted to examine associations between categorical variables and seasonality. One-way ANOVAs were conducted to compare continuous variables across seasonality groups. A chi-

square goodness-of-fit test was used to compare the number of concussions sustained across the 4 parts of the season. To compare SR and RTP across seasonality (preseason, early, middle, and late), Kaplan-Meier survival curves were generated for each seasonality group. Differences in SR and RTP across seasonality groups were assessed using the log-rank test. To assess the predictive value of seasonality on SR and RTP, Cox proportional hazards regression analyses were conducted. Seasonality was the main independent variable; preseason, early, middle, and late-season were dummy coded in regression analyses. Covariates included age (continuous), days from injury to post-injury SCAT3 testing (continuous), post-injury SCAT3 TSS assessed within 24 hours (continuous), altered mental status (0 = no, 1 = yes), loss of consciousness (0 = no, 1 = yes), amnesia (0 = no, 1 = yes), LD (0 = no, 1 = yes), ADHD (0 = no, 1 = yes), and psychological disorders (0 = no, 1 = yes). Covariates were selected to isolate the impact of seasonality on recovery and adjust for known predictors of concussion outcomes.[4-12] Statistical significance for all tests was set *a priori* at $\alpha = .05$. All analyses were performed using IBM SPSS Statistics 27 (IBM, Armonk, NY).

RESULTS

A total of 1,204 football players sustained a concussion. Out of these athletes, 56 were excluded due to non-sport-related concussions, 261 were excluded due to non-Division I status, and 205 were excluded due to sustaining a concussion outside of the defined study period. The final sample included 682 preseason and regular-season concussions. The average age of participants (in years) differed significantly by seasonality group [$F(3677) = 3.216$, $P = .022$], while all other demographic, medical history, and injury characteristics variables were similar across groups (Table 1).

Concussion incidence

Of the 682 regular-season concussions, the incidence rate was highest during preseason (4.73 per 100 players) compared to early (2.19 per 100 players), middle (2.31 per 100 players), and late-season (2.17 per 100 players). In terms of temporal distribution, 286 (41.9%) occurred during the preseason, 131 (19.2%) during the early-season, 137 (20.1%) during the middle season, and 128 (18.8%) during the late-season (Figure 1). There was a significant difference in the frequency of concussion between the 4 parts of the season, compared to expected frequencies ($\chi^2 = 104.6$, $P < .001$). While each team/season has a specific set of dates that were divided to comprise preseason, early, middle, and late-season, overall, "preseason" tended to constitute August 1 to August 31, "early-season" was typically from September 1 to September 29, "middle season" was typically from September 30 to October 28, and "late-season" was typically from October 29 to November 27.

Time to SR

Kaplan-Meier analysis showed significant differences in time to SR across preseason (median: 5.7 days, 95% CI: 5.1-6.3), early-season (median: 6.6 days, 95% CI: 5.1-8.1), middle season (median: 4.8 days, 95% CI: 4.0-5.6), and late-season (median: 6.9 days, 95% CI: 5.5-8.3) (log-rank $P = .012$) (Figure 2). Cox proportional hazards regression revealed that early-season concussions were associated with a lower hazard of SR (HR = 0.582, 95% CI: 0.429-0.788, $P < .001$) compared to preseason concussions, indicating a 41.8% reduced likelihood of SR at any given time point during the follow-up period. Similarly, late-season concussions were associated with delayed SR (HR = 0.634, 95% CI: 0.473-0.850, $P = .002$) compared to preseason concussions, indicating a 35.6% reduced likelihood of SR at any given time point during the follow-up period (Table 2). Other predictors of longer SR included increased days from injury to SCAT3 testing (HR = 0.810, 95% CI: 0.742-0.885, $P < .001$), higher post-injury SCAT3 TSS (HR = 0.902, 95% CI: 0.884-0.920, $P < .001$), and absence of loss of consciousness (HR = 0.539, 95% CI: 0.307-0.946, $P = .031$) (Table 2).

Time to RTP

Kaplan-Meier analysis showed no significant differences in time to RTP across preseason (median: 10.7 days, 95% CI: 9.7-11.7), early-season (median: 10.7 days, 95% CI: 9.7-11.7), middle season (median: 11.1 days, 95% CI: 10.2-12.0), and late-season (median: 12.7 days, 95% CI: 9.9-15.5) (log-rank $P = .300$) (Figure 3). Cox regression demonstrated that early-season concussions were associated with a lower hazard of RTP (HR = 0.703, 95% CI: 0.502-0.983, $P = .040$) compared to preseason concussions, indicating a 29.7% reduced likelihood of RTP at any given time point during the follow-up period (Table 2). Increased days from injury to SCAT3 testing (HR = 0.896, 95% CI: 0.832-0.965, $P = .004$) and higher post-injury SCAT3 TSS (HR = 0.931, 95% CI: 0.912-0.950, $P < .001$) were also associated with longer RTP (Table 2).

DISCUSSION

The current study investigated the effect of seasonality on concussion incidence and recovery, defined as time to SR and RTP, in Division I college football athletes. Out of the 682 concussions that occurred from the start of preseason to the end of the regular-season, incidence was highest during preseason compared to early, middle, and late-season. Regarding time to recovery, both early and

TABLE 1 Demographics, medical history, and injury characteristics

	Entire sample (*n* = 682)	Preseason (*n* = 286)	Early season (*n* = 131)	Middle season (*n* = 137)	Late season (*n* = 128)	Statistics	*P*
Demographics							
Age (M ± SD)	20.4 (1.3)	20.2 (1.2)	20.4 (1.3)	20.5 (1.3)	20.6 (1.4)	F(3677) = 3.216	**.022**
Race (*n*, %)							
White	237 (34.8%)	96 (33.6%)	47 (35.9%)	47 (34.3%)	47 (36.7%)	x^2 = 12.742	.807
Black	277 (40.6%)	121 (42.3%)	53 (40.5%)	58 (42.3%)	45 (35.2%)		
Multiple races	51 (7.5%)	22 (7.7%)	10 (7.6%)	8 (5.8%)	11 (8.6%)		
Other/unknown	117 (17.2%)	47 (16.4%)	21 (16.0%)	24 (17.5%)	25 (19.5%)		
Ethnicity							
Non-Hispanic/Latino	397 (58.2%)	164 (57.3%)	74 (56.5%)	75 (54.7%)	84 (65.6%)	x^2 = 15.066	.238
Medical history							
ADHD (n, %)	57 (8.4%)	27 (9.4%)	9 (6.9%)	6 (4.4%)	15 (11.7%)	x^2 = 5.118	.163
Learning disabilities (%)	24 (3.5%)	7 (2.4%)	4 (3.1%)	7 (5.1%)	6 (4.7%)	x^2 = 2.738	.434
Migraine history (%)	43 (6.3%)	22 (7.7%)	9 (6.9%)	6 (4.4%)	6 (4.7%)	x^2 = 2.428	.488
Non-migraine headache history (%)	15 (2.2%)	6 (2.1%)	3 (2.3%)	2 (1.5%)	4 (3.1%)	x^2 = 0.812	.847
Psychological Disorder (%)	13 (1.9%)	4 (1.4%)	3 (2.3%)	2 (1.5%)	4 (3.1%)	x^2 = 1.605	.658
Baseline SCAT3 TSS (M ± SD)	3.8 (6.9)	4.1 (7.5)	3.9 (6.3)	3.4 (6.4)	3.3 (6.5)	F(3559) = 0.445	.721
Injury characteristics							
Altered mental status (%)	357 (52.3%)	158 (55.2%)	69 (52.7%)	69 (50.4%)	61 (47.7%)	x^2 = 1.802	.615
Loss of consciousness (%)	30 (4.4%)	11 (3.8%)	5 (3.8%)	4 (2.9%)	10 (7.8%)	x^2 = 4.581	.205
Amnesia (%)	94 (13.8%)	35 (12.2%)	24 (18.3%)	19 (13.9%)	16 (12.5%)	x^2 = 3.081	.378
Days from injury to SCAT3[a] (M ± SD)	1.5 (1.4)	1.5 (1.1)	1.6 (2.0)	1.6 (1.5)	1.3 (1.4)	F(3526) = 0.902	.440
SCAT3 TSS (M ± SD)	27.9 (23.1)	29.2 (23.5)	25.5 (24.0)	26.0 (22.3)	29.7 (22.0)	F(3546) = 1.071	.361
Days to symptom resolution (M ± SD)	7.8 (7.6)	6.6 (5.2)	9.0 (7.9)	7.5 (8.2)	8.8 (10.2)	F(3568) = 3.421	**.017**
Day to return-to-play (M ± SD)	13.0 (8.5)	12.3 (7.2)	13.5 (9.7)	14.2 (10.9)	13.0 (7.6)	F(3492) = 1.166	.322

Abbreviations: ADHD, Attention-deficit/Hyperactivity disorder; TSS, total symptom scale.
[a]Days from injury to SCAT3 testing.
Bold values indicate statistical significance (P<.05).

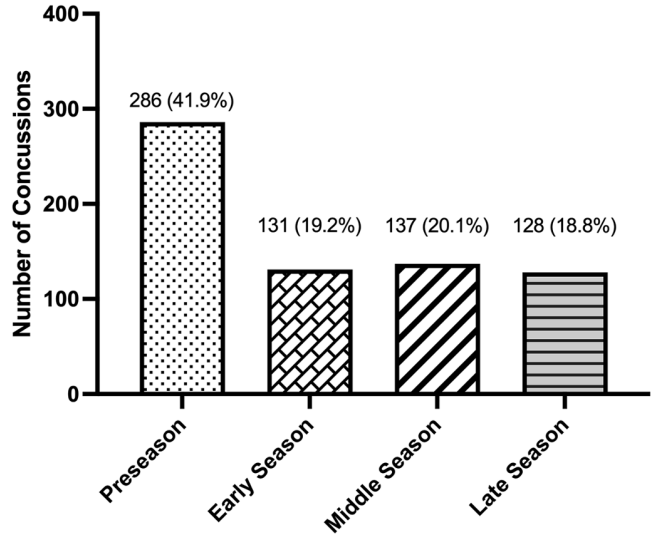

Figure 1. Frequency of concussions across preseason, early, middle, and late-season.

late-season concussions were associated with a 42% and 37% lower hazard of SR, respectively, compared to preseason concussions. Early-season concussions were also associated with a 30% lower hazard of RTP compared to preseason concussions. These hazard-based findings corresponded to median SR durations ranging from 4.8 to 6.9 days, and median RTP durations ranging from 10.7 to 12.7 days. While absolute differences were modest, the consistent pattern across models indicates that seasonality is associated with recovery timelines, warranting further investigation into underlying mechanisms. The current study establishes these temporal associations but does not evaluate whether seasonality provides independent prognostic value beyond known predictors such as symptom severity, prior concussion history, or loss of consciousness.

Concussion incidence was highest during the football preseason, consistent with findings from previous studies.[17,18] For instance, in a study examining 19 NCAA Division I football programs, concussions

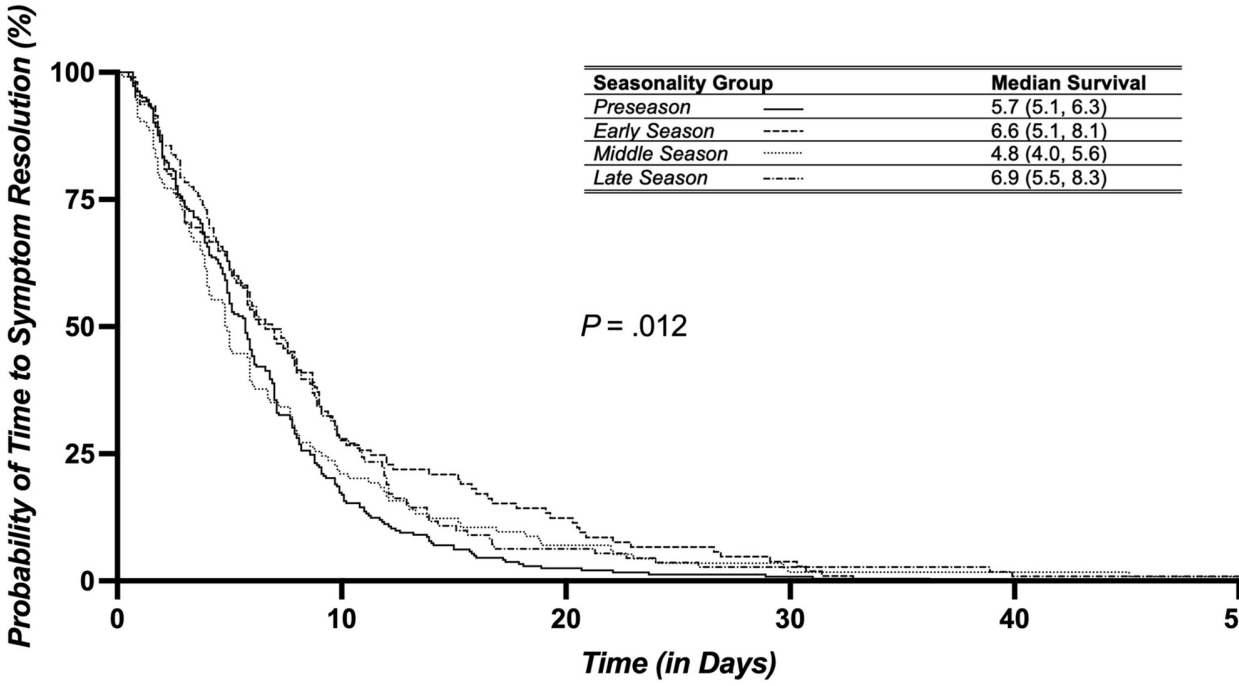

Figure 2. Survival curve for time to symptom resolution.

TABLE 2 *Cox hazard regression predicting days to symptom resolution and return-to-play*

	β	SE	Wald	HR	95% CI	*P*
Symptom resolution						
Early-season	−0.542	0.155	12.199	0.582	0.429, 0.788	**<.001**
Middle season	−0.106	0.154	0.470	0.900	0.665, 1.217	.493
Late-season	−0.456	0.150	9.305	0.634	0.473, 0.850	**.002**
Age	−0.009	0.044	0.039	0.991	0.910, 1.080	.844
Days from injury to SCAT3 TSS	−0.210	0.045	21.728	0.810	0.742, 0.885	**<.001**
SCAT3 TSS	−0.103	0.010	105.673	0.902	0.884, 0.920	**<.001**
Altered mental status	0.078	0.116	0.443	1.081	0.860, 1.358	.506
Loss of consciousness	0.618	0.287	4.641	1.856	1.057, 3.256	.031
Amnesia	−0.121	0.164	0.542	0.886	0.643, 1.222	.462
Learning disorder	−0.248	0.261	0.901	0.781	0.468, 1.302	.343
ADHD	0.160	0.179	0.804	1.174	0.827, 1.666	.370
Psychological disorder	0.071	0.361	0.038	1.073	0.529, 2.179	.845
Return-to-play						
Early-season	−0.353	0.172	4.232	0.703	0.502, 0.983	**.040**
Middle season	−0.299	0.168	3.185	0.742	0.534, 1.030	.074
Late-season	−0.230	0.160	2.063	0.794	0.580, 1.088	.151
Age	0.007	0.046	0.026	1.007	0.921, 1.102	.871
Days from injury to SCAT3 TSS	−0.110	0.038	8.505	0.896	0.832, 0.965	**.004**
SCAT3 TSS	−0.072	0.010	48.040	0.931	0.912, 0.950	**<.001**
Altered mental status	−0.008	0.125	0.004	0.992	0.776, 1.269	.951
Loss of consciousness	0.121	0.376	0.104	1.129	0.540, 2.358	.747
Amnesia	−0.196	0.176	1.237	0.822	0.583, 1.161	.266
Learning disorder	0.249	0.338	0.541	1.283	0.661, 2.490	.462
ADHD	0.130	0.197	0.434	1.138	0.774, 1.674	.510
Psychological disorder	0.451	0.424	1.131	1.569	0.684, 3.600	.288

Abbreviations: ADHD, attention deficit/hyperactive disorder; TSS, total symptom scale.
Bold values indicate statistical significance (P<.05).

occurred most frequently during the spring and preseason periods.[17] Similarly, in a study of 6 NCAA Division I college football programs, the preseason accounted for 48.5% of total concussions, even though it represented only 20% of the total season time.[18] These results offer a different perspective on the temporal distribution of sport-related concussions in collegiate football, diverging from studies in high school football, where middle-season concussions (39.2%) were most prevalent,[14] and professional American football, which reported a higher frequency of concussions in the second half of the regular-season.[23] The prevalence of preseason concussions prompts important questions about the underlying factors contributing to this trend and its potential implications for athlete safety and injury prevention. Several factors may contribute to the higher incidence of preseason concussions. First, prior research has shown that collegiate American football players experience a greater than 40% increase in recorded head impacts in the preseason compared to the regular-season,[24] which naturally increases the likelihood of concussion.[25] Similarly, preseason often involves intense training sessions and scrimmages designed to prepare players for the regular-season.[26] These high-intensity practices might lead to more aggressive play and higher risk-taking behaviors, especially in fully padded practices.[25] The pressure to perform well during preseason to secure starting positions or playing time might also push players to exceed their current physical capabilities,[27] further elevating the risk of injury. Players may also be deconditioned or not fully in game shape during the preseason,[28] which can result in improper form and increased injury risk. Furthermore, teams may not yet be fully synchronized in executing plays, leading to errors on the field and a higher chance of injury. Future research exploring the interplay of these elements and their impact on concussion patterns during different phases of the season could offer a more comprehensive understanding of the observed frequency distribution. Importantly, these explanations remain speculative. The current observational design cannot distinguish among potential mechanisms such as increased head impact exposure, training intensity, roster competition pressures, or deconditioning. Future studies incorporating direct measurement of these factors (eg, head impact telemetry, roster status, and psychological assessments) are needed to test these hypotheses.

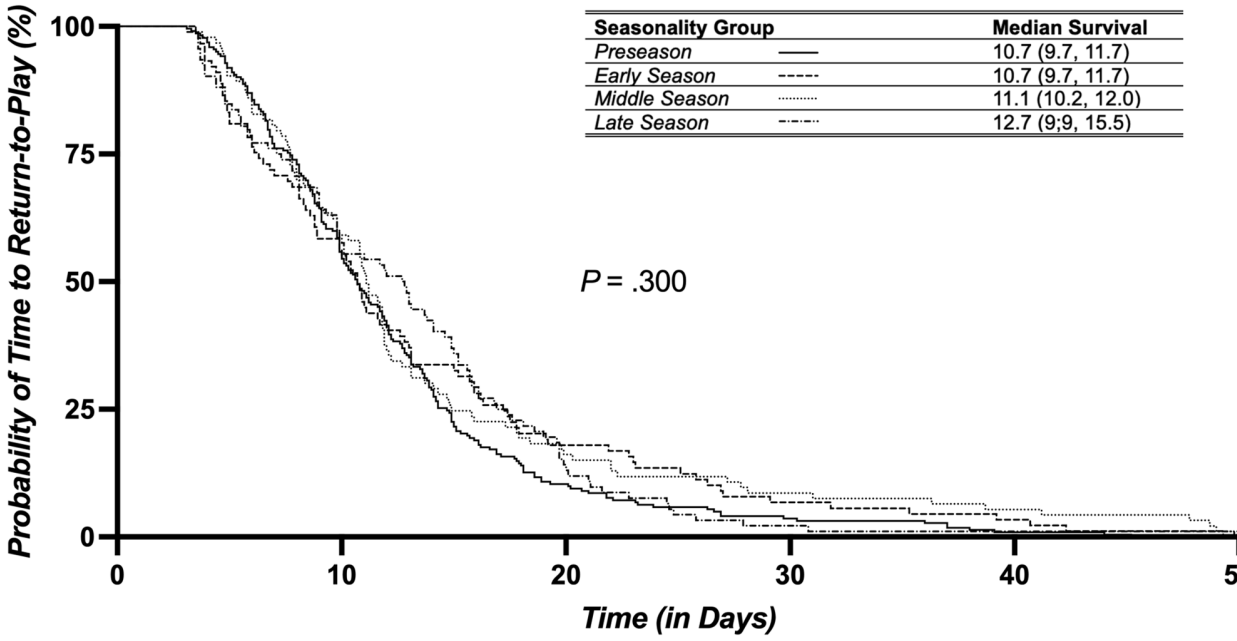

Figure 3. Survival curve for time to return-to-play.

Both early and late-season concussions exhibited significantly longer SR compared to preseason concussions, while early-season concussions exhibited significantly longer RTP compared to preseason. Although the absolute differences in median SR and RTP were modest and may not be clinically meaningful in isolation, the hazard findings from Cox regression offer a more complete picture by evaluating recovery probabilities across the full follow-up period. These results provide a nuanced picture of concussion recovery in collegiate football players and differ from some previous research. A recent study of high school football athletes reported that late-season concussions had a longer time to SR (25.5 ± 27.0 days) compared to their early-season counterparts (11.5 ± 12.9 days).[14] However, no differences were seen regarding time to return-to-learn or RTP.[14] Another recent study examining a variety of high school sports found no discernible impact of seasonality on RTL, SR, or RTP.[15] The differences in findings may be due to differences in samples, as prior studies derived their samples from specialty concussion clinics, while the present study prospectively investigates all college football players from a variety of football programs. This prospective dataset may have less sampling bias than previous investigations on this topic. The smaller magnitude of group differences in our cohort may also reflect more standardized RTP procedures at the collegiate level, reducing variability in recovery timelines.

Several factors may contribute to the observed recovery patterns, although our study design precludes definitive conclusions about underlying mechanisms. The shorter SR and RTP for preseason concussions may be facilitated by the greater flexibility in scheduling (pre-academic coursework and less intensive travel),[29] and the absence of cumulative physical or head impact exposure from prior practices/games, both of which support adherence to recovery protocols and more favorable recovery trajectories.[30] Conversely, the longer SR and RTP times for early-season concussions may be linked to the sharp transition into the competitive season.[31] Moreover, athletes and athletic trainers may exercise greater caution given the long season ahead, avoiding rushed RTP decisions. The prolonged SR seen in late-season concussions may be influenced by cumulative fatigue and the physical toll of the season.[32] Alternatively, late-season injuries may occur when remaining games are insufficient for full return, or when playoff eligibility is no longer viable. It is important to note that statistical significance may not directly translate to clinical significance. While hazard ratios indicate relative differences in recovery timing based, absolute differences in median days should guide clinical management decisions. Importantly, these proposed mechanisms remain speculative. Our observational design cannot distinguish among potential explanatory factors such as training load differences, psychological pressures, cumulative repetitive head impacts, academic demands, or adherence variations. Future studies incorporating direct measurement of these factors are essential to understanding why these temporal patterns emerge.

Implications

Our findings establish that temporal patterns in both incidence and recovery exist across season phases.

While absolute differences are modest (1-2 days for SR) and may not warrant season-specific protocols, the consistency of these patterns indicates that contextual factors systematically influence outcomes and warrant mechanistic investigation. Seasonality may serve as a marker for contextual pressures or cumulative burden of head impacts. Athletic trainers should recognize that preseason concussions (which showed faster resolution) may coincide with roster competition pressures, while early-season injuries overlap with academic demands and late-season injuries occur amid playoff stakes and cumulative fatigue. Alternatively, higher preseason concussion frequency may reflect more comprehensive medical surveillance, greater athlete willingness to disclose mild symptoms when not yet competing in high-stakes regular-season games, or the effects of deconditioning from the off-season. The elevated preseason incidence suggests that targeted prevention and surveillance during this period may be beneficial. These findings are meant to be hypothesis-generating. Whether seasonality adds prognostic utility beyond established clinical predictors requires future investigation. If specific mechanisms such as deconditioning or cumulative exposure are confirmed through direct measurement, targeted primary prevention interventions could be developed. For example, testing whether roster status (starters vs. prospects) moderates the relationship between seasonality and recovery would directly examine competitive pressure as a mechanism. Similarly, incorporating head impact telemetry data could clarify whether cumulative burden of repetitive head impacts explains late-season recovery delays. Future research measuring these factors directly is essential to understanding why these patterns emerge and to developing targeted interventions.

Limitations

Several limitations warrant acknowledgment. The use of relatively arbitrary cutoffs regarding seasonality and absence of a standardized definitions for early, middle, and late-seasons may introduce variability in results across studies, underscoring the imperative need for a more uniform classification system. It is crucial to highlight that the current study's definitions of early, middle, and late-season concussions diverge from those employed in prior studies. While this approach offers the advantage of excluding playoffs from the designated "season" period, given that not all teams advance to the playoffs, it is essential to recognize that these operational definitions may impede direct comparisons with studies utilizing different temporal categorizations. Although dividing the regular-season into 3 discrete time epochs facilitated consistent comparisons across teams and seasons, this approach inherently involves some limitations. Specifically, cases occurring near cutoff points may be classified into adjacent epochs based on a single-day difference, potentially reducing the granularity of seasonal context. While methodologically practical, this framework may not fully capture the continuous nature of risk variation over time. Furthermore, the exclusion of concussions occurring outside the defined study period may curtail the generalizability of findings to the broader context of football seasons. Lack of exposure data prevents accurate incidence rate calculations, as it is unclear whether all "at-risk" athletes had equal exposure throughout the season, aside from time missed due to concussion recovery. The study's focus on college football athletes also limits the generalizability of findings to other sports, competition levels, or age groups. Replication in other sports and at different competitive levels is necessary to determine whether seasonality effects represent a sport-specific phenomenon or a more universal pattern across athletic contexts. Given the multifaceted nature of concussions, influenced by various individual and environmental factors, a more comprehensive investigation into the specific mechanisms underlying seasonality effects is warranted. Finally, it is important to note that concussion identification and management evolved over the study period (2014-2020), which may have influenced the incidence, reporting, and recovery patterns observed in this cohort.

CONCLUSIONS

In this large, prospective cohort study from 682 Division I collegiate football athletes, preseason concussions had the highest incidence. Compared to preseason concussions, early and late-season concussions were associated with longer symptom resolution, while early-season concussions were associated with longer time to return-to-play. These results highlight the nuanced impact of seasonality on incidence, symptom resolution and RTP, particularly for college football players who face a rigorous schedule.

REFERENCES

1. Gessel LM, Fields SK, Collins CL, Dick RW, Comstock RD. Concussions among United States high school and collegiate athletes. *J Athl Train*. 2007;42(4):495–503.
2. Pierpoint LA, Collins C. Epidemiology of sport-related concussion. *Clin Sports Med*. 2021;40(1):1–18. doi:10.1016/j.csm.2020.08.013.
3. Zuckerman SL, Kerr ZY, Yengo-Kahn A, Wasserman E, Covassin T, Solomon GS. Epidemiology of sports-related concussion in NCAA athletes from 2009-2010 to 2013-2014: incidence, Recurrence, and mechanisms. *Am J Sports Med*. 2015;43 (11):2654–2662. doi:10.1177/0363546515599634.

4. Zuckerman SL, Totten DJ, Rubel KE, Kuhn AW, Yengo-Kahn AM, Solomon GS. Mechanisms of injury as a diagnostic predictor of sport-related concussion severity in football, basketball, and soccer: results from a regional concussion registry. *Neurosurgery*. 2016;63 Suppl 1(Supplement 1):102–112. doi:10.1227/NEU.0000000000001280.
5. Terry DP, Huebschmann NA, Maxwell BA, et al. Preinjury migraine history as a risk factor for prolonged return to school and sports following concussion. *J Neurotrauma*. 2019;36(1):142–151. doi:10.1089/neu.2017.5443.
6. Scott BR, Uomoto JM, Barry ES. Impact of pre-existing migraine and other co-morbid or co-occurring conditions on presentation and clinical course following deployment-related concussion. *Headache*. 2020;60(3):526–541. doi:10.1111/head.13709.
7. Sandel NK, Schatz P, Goldberg KB, Lazar M. Sex-based differences in cognitive deficits and symptom reporting among acutely concussed adolescent lacrosse and soccer players. *Am J Sports Med*. 2017;45(4):937–944. doi:10.1177/0363546516677246.
8. Gaudet CE, Iverson GL, Zafonte R, Berkner PD, Cook NE. Prior concussion history and clinical recovery after sport-related concussion in high school athletes. *J Neurotrauma*. 2023;40(13-14):1459–1469. doi:10.1089/neu.2022.0358.
9. Chizuk HM, Cunningham A, Horn EC, et al. Association of concussion history and prolonged recovery in youth. *Clin J Sport Med Off J Can Acad Sport Med*. 2022;32(6):e573–e579. doi:10.1097/JSM.0000000000001044
10. Kuhn AW, Zuckerman SL, Yengo-Kahn AM, et al. Factors associated with playing through a sport-related concussion. *Neurosurgery*. 2017;64(CN_suppl_1):211. doi:10.1093/neuros/nyx294
11. Kerr ZY, Register-Mihalik JK, Kroshus E, Baugh CM, Marshall SW. Motivations associated with nondisclosure of self-reported concussions in former collegiate athletes. *Am J Sports Med*. 2016;44(1):220–225. doi:10.1177/0363546515612082.
12. Dompier TP, Kerr ZY, Marshall SW, et al. Incidence of concussion during practice and games in youth, high school, and collegiate american football players. *JAMA Pediatr*. 2015;169(7):659–665. doi:10.1001/jamapediatrics.2015.0210
13. Brett BL, Kuhn AW, Yengo-Kahn AM, et al. Initial symptom presentation after high school football-related concussion varies by time point in a season: an initial investigation. *Sports Med - Open*. 2018;4(1):8. doi:10.1186/s40798-018-0121-8
14. Jo J, Hill TM, Williams KL, et al. Do sport-related concussions later in the season take longer to recover from? *Clin J Sport Med Off J Can Acad Sport Med*. 2024;34(2):97–104. doi:10.1097/JSM.0000000000001189.
15. Bishay AE, Williams KL, Jo J, Fitch S, Terry DP, Zuckerman SL. Does recovery after sport-related concussion vary by time point in a season? A multi-sport investigation. *J Neurosurg Pediatr*. 2023;33(3):268–275. doi:10.3171/2023.10.PEDS23392
16. Stemper BD, Harezlak J, Shah AS, et al. Association between preseason/regular season head impact exposure and concussion incidence in NCAA football. *Med Sci Sports Exerc*. 2022;54(6):912–922. doi:10.1249/MSS.0000000000002874
17. Broglio SP, Perkins SM, Riggen L, et al. Similar concussion rates in spring football and preseason: findings from the concussion assessment, research and education consortium. *J Athl Train*. 2023;58(3):220–223. doi:10.4085/1062-6050-0132.22
18. McCrea MA, Shah A, Duma S, et al. Opportunities for prevention of concussion and repetitive head impact exposure in college football players: a concussion assessment, Research, and Education (CARE) consortium study. *JAMA Neurol*. 2021;78(3):346–350. doi:10.1001/jamaneurol.2020.5193
19. How the 12-team college football playoff will work: teams, schedule, bids | NCAA.com. Accessed December 20, 2023. https://www.ncaa.com/news/football/article/2023-12-03/how-12-team-college-football-playoff-will-work-teams-schedule-bids
20. Broglio SP, McCrea M, McAllister T, et al. A national study on the effects of concussion in collegiate athletes and US military service academy members: the NCAA-DoD Concussion Assessment, Research and Education (CARE) consortium structure and methods. *Sports Med Auckl NZ*. 2017;47(7):1437–1451. doi:10.1007/s40279-017-0707-1
21. Carney N, Ghajar J, Jagoda A, et al. Concussion guidelines step 1: systematic review of prevalent indicators. *Neurosurgery*. 2014;75 Suppl 1(Supplement 1):S3–15. doi:10.1227/NEU.0000000000000433
22. Knowles SB, Marshall SW, Guskiewicz KM. Issues in estimating risks and rates in sports injury research. *J Athl Train*. 2006;41(2):207–215.
23. Dai JB, Li AY, Haider SF, et al. Effects of game characteristics and player positions on concussion incidence and severity in professional football. *Orthop J Sports Med*. 2018. 6(12). doi:10.1177/2325967118815448
24. Stemper BD, Shah AS, Harezlak J, et al. Repetitive head impact exposure in college football following an NCAA rule change to eliminate two-a-day preseason practices: a study from the NCAA-DoD CARE consortium. *Ann Biomed Eng*. 2019;47(10):2073–2085. doi:10.1007/s10439-019-02335-9
25. Wasserman EB, Coberley M, Anderson S, Grant M, Hardin JA. Concussion rates differ by practice type and equipment worn in an autonomy five collegiate football conference. *Clin J Sport Med Off J Can Acad Sport Med*. 2020;30(4):366–371. doi:10.1097/JSM.0000000000000615.
26. Sampson JA, Murray A, Williams S, et al. Injury risk-workload associations in NCAA American college football. *J Sci Med Sport*. 2018;21(12):1215–1220. doi:10.1016/j.jsams.2018.05.019
27. Li H, Moreland JJ, Peek-Asa C, Yang J. Preseason anxiety and depressive symptoms and prospective injury risk in collegiate athletes. *Am J Sports Med*. 2017;45(9):2148–2155. doi:10.1177/0363546517702847.
28. Hoffman JR, Ratamess NA, Klatt M, et al. Comparison between different off-season resistance training programs in Division III American college football players. *J Strength Cond Res*. 2009;23(1):11–19. doi:10.1519/jsc.0b013e3181876a78
29. Division I football oversight committee recommends preseason changes. *NCAA.org*. May 6, 2021. Accessed August 15, 2024. https://www.ncaa.org/news/2021/5/6/division-i-football-oversight-committee-recommends-preseason-changes.aspx
30. Roberts AL, Pascual-Leone A, Speizer FE, et al. Exposure to American football and neuropsychiatric health in former National Football League players: findings from the football players health study. *Am J Sports Med*. 2019;47(12):2871–2880. doi:10.1177/0363546519868989
31. Hamlin MJ, Wilkes D, Elliot CA, Lizamore CA, Kathiravel Y. Monitoring training loads and perceived stress in young elite university athletes. *Front Physiol*. 2019;10:34. doi:10.3389/fphys.2019.00034.
32. Jones CM, Griffiths PC, Mellalieu SD. Training load and fatigue marker associations with injury and illness: a systematic review of longitudinal studies. *Sports Med Auckl Nz*. 2017;47(5):943–974. doi:10.1007/s40279-016-0619-5.